Temporal Score Incomparability in LLM Benchmarks: A Large-Scale Psychometric Audit of 5,227 Models

Anonymous ACL submission

Abstract

LLM benchmark scores are compared across model generations, yet whether individual items measure the same construct over time is rarely tested. We apply differential item functioning (DIF) analysis from educational measurement to six METABENCH benchmarks (5,227 models), validated through a semi-synthetic injection framework with cross-subject matching. Between 2023 and 2024 model cohorts, 17.9–31.3% of items exhibit large temporal DIF (29.2% for MMLU after non-independence correction, cluster bootstrap 95% CI: $[26.7\%, 53.2\%]$; $>45\times$ random baseline). Web-overlap labels (a proxy for training-set contamination) are near-independent of DIF flags: global exposure and differential functioning are complementary diagnostic dimensions. Temporal DIF does not threaten aggregate rankings ($\rho \approx 0.997$) but reveals structured item-level change—base and instruct models exhibit opposite DIF patterns ($\chi^2 = 562.7$), evidence that DIF captures capability shifts beyond contamination—and shifts DIF-corrected scores by $+2.12$ pp on average under full C-flag removal. We propose a three-step audit protocol (Monitor/Triage/Correct) with empirically grounded thresholds for benchmark maintainers. LLM benchmarks need psychometric monitoring, not just score reporting.

1 Introduction

When we say "Model B outperforms Model A on MMLU," we implicitly assume that the benchmark measures the same construct in the same way for both models. This assumption of temporal comparability underpins every cross-generation comparison published on leaderboards and in model reports, yet it is rarely tested. We use *measurement invariance* to denote the property that test items function identically across groups (Meredith, 1993), *differential item functioning* (DIF) as the item-level statistical test for violations, and *temporal score incomparability* (measurement non-invariance across temporal cohorts, manifesting as score incomparability) as the empirical phenomenon when DIF is detected—scores are not directly comparable across cohorts because item measurement properties differ systematically, without implying degradation or attributing cause. Applying Mantel-Haenszel DIF analysis (Holland and Thayer, 1988) with ETS severity classification to the METABENCH response matrix (Kipnis et al., 2024),[1] we find that 31% of MMLU (Hendrycks et al., 2021) items receive ETS Category C classification ($|\Delta_{\mathrm{MH}}| \geq 1.5$, $p < 0.05$) between 2023 and 2024 model cohorts—versus 0.62% under a random-split control. Aggregate rankings are robust ($\rho \approx 0.997$), but absolute scores are not directly comparable: DIF-corrected scoring (excluding all C-flagged items) shifts accuracy by $+2.12$ pp on average, with individual models displaced by up to 849 rank positions. This instability does not invalidate the benchmark—aggregate rankings are massively redundant—but it is diagnostic, revealing which capability dimensions shift across model generations.

Existing approaches to benchmark integrity focus on *global exposure*—whether items appear in training corpora—through membership inference (Shi et al., 2024) and performance-based detection (Dekoninck et al., 2024; Liang, 2026; Tao et al., 2026), validated against web-overlap labels (Li et al., 2024b). Whether benchmark items function equivalently across model generations—*measurement invariance*—has not been systematically examined (Figure 1). We find these two signals are near-independent (Jaccard $= 0.206$, $\phi = 0.012$): complementary diagnostic dimensions, not competing ones.

We apply a temporal DIF audit framework to the

[1] Analysis code and DIF audit scripts are available at https://github.com/anonymous-nlp-2026-2/irt_equated_bench_regen.

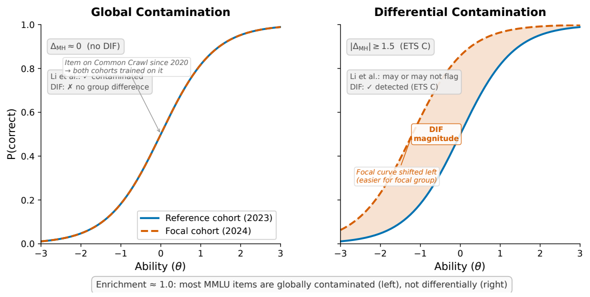

Figure 1: Schematic illustration of global vs. differential contamination via Item Characteristic Curves (ICCs). **Left**: global contamination produces no DIF; both cohorts encounter the item, and their ICCs overlap. **Right**: differential contamination produces DIF; the focal cohort's ICC shifts because only that cohort was disproportionately exposed during training (assuming no ceiling effect). The enrichment ratio of ≈ 1.0 between DIF flags and web-overlap labels is expected because most MMLU items fall into the left category.

full METABENCH response matrix (5,227 models $\times$ 12,508 MMLU items), to our knowledge the largest-scale DIF analysis on LLM benchmarks, complementing concurrent work on anchor-item calibration (Habba et al., 2026) and addressing the fragility of existing detectors under post-training perturbation (Wang et al., 2026). Two structural controls, a sensitivity check, and replication across all six METABENCH benchmarks confirm that the temporal DIF rate (29.2% after non-independence correction; 17.9–31.3% across benchmarks) is not a methodological artifact (Sections 4.2, 4.4 and 4.7).

Our contributions are:

1. **Temporal DIF at scale, complementary to contamination detection**: We establish that 29.2% of MMLU items exhibit temporal DIF across 2023–2024 model cohorts after non-independence correction (cluster bootstrap 95% CI: $[26.7\%, 53.2\%]$; 17.9–31.3% across six benchmarks)—a signal near-independent of web-overlap labels (Jaccard $=$ 0.206). Base-vs-instruct stratification reveals opposite DIF patterns ($\chi^2 = 562.7$), providing label-independent evidence for mechanism separation. DIF-corrected scoring shifts accuracy by $+2.12$ pp on average, with individual models displaced by up to 849 positions (Sections 4.2, 4.3, 4.6, 4.7 and 5.1).

2. **Cross-subject matching with constructive FPR guarantee**: A cross-subject matching strategy provides a constructive $\Delta\text{FPR} = 0$ guarantee under single-subject contamination, extending empirically to $k \leq 5$, solving matching-variable corruption when the benchmark itself may be compromised (Section 4.5). We complement this with family-level DEFF correction for non-independence and semi-synthetic validation for detection-power calibration.

3. **Actionable audit protocol**: We translate these findings into a three-step protocol (Monitor/Triage/Correct) with empirically grounded thresholds for benchmark maintainers (Section 5.2).

2 Related Work

DIF Methodology and Measurement Invariance. The Mantel-Haenszel procedure for DIF detection was introduced by Holland and Thayer (1988) and remains the most widely used method in educational testing due to its simplicity and well-characterized statistical properties (Clauser and Mazor, 1998). Alternative approaches include logistic regression DIF (Swaminathan and Rogers, 1990), SIBTEST (Shealy and Stout, 1993), Lord's χ^2 test (Lord, 1980), and the Raju area method (Raju, 1988); Millsap and Everson (1993), Zumbo (1999), and Penfield and Camilli (2007) provide comprehensive reviews. DIF analysis is a specific instance of measurement invariance testing (Meredith, 1993; Vandenberg and Lance, 2000; Millsap, 2011), which examines whether a measurement instrument functions equivalently across populations. MIMIC models offer a latent-variable approach to the same problem (Muthén, 1985; Woods, 2009), and Teresi and Fleishman (2007) reviews DIF methods across applied domains. Cross-subject matching—computing the stratification variable from domains other than the one being tested—adapts purified subtest matching from the DIF literature (Nandakumar, 1993; Donoghue and Allen, 1993) to a new challenge: contamination control at scale, where the matching variable itself may be compromised by the phenomenon under test. We retain the standard MH statistic and develop LLM-specific adaptations around it: cross-subject matching for contamination control (with a constructive $\Delta\text{FPR} = 0$ guarantee), DEFF correction for non-independence, and semi-synthetic validation absent ground-truth labels. The 5,227-examinee scale exceeds typical DIF applications (Belzak, 2020). The impact-versus-DIF confound is addressed through matched-ability quintile analysis, IRT θ-conditioning (as sensitivity check), and a random-split control (Section 4.2).

IRT in LLM Evaluation. IRT adoption in LLM evaluation has accelerated: Lalor et al. (2019) introduced IRT to NLP, Vania et al. (2021) assessed test set discriminability, and Rodriguez et al. (2021) demonstrated that not all evaluation examples are equally informative. Recent extensions include PSN-IRT (Zhou et al., 2026) for benchmark quality at scale, TinyBenchmarks (Maia Polo et al., 2024) for IRT-based item selection, scalability coefficients for detecting bad benchmark items (Hardy et al., 2026), IRT-based evaluation quality (Kraus and Arora, 2024), ATLAS (Li et al., 2025) for computerized adaptive testing of LLMs, JE-IRT (Yao et al., 2025) for geometric ability embedding, continuous-score adaptive evaluation (Balkır et al., 2026), and competency-focused IRT evaluation in medical domains (Luo et al., 2025). The closest concurrent work is Growing Pains (Habba et al., 2026), which uses IRT calibration to establish anchor items for cross-temporal score equating via fixed-parameter calibration (FPC; Kim and Kolen, 2019). While Growing Pains uses parametric IRT (2PL/3PL fixed-parameter calibration), MH-DIF is nonparametric and robust to model misspecification—an advantage given poor 2PL fit for MMLU (RMSEA > 0.08; Section A.2). Growing Pains prescribes how to equate scores after instability is detected; our work diagnoses *which* items exhibit instability and quantifies severity via ETS classification, providing pre-screened anchor candidates for downstream equating. Their anchor-item selection can directly consume our stable-item set as pre-screened candidates; our finding that 31% of items exhibit temporal DIF identifies which items should *not* serve as anchors. To our knowledge, no prior work applies DIF to benchmark temporal comparability—contamination detection and psychometric evaluation optimization have developed as separate threads.

Contamination Detection. A growing family of methods detects benchmark contamination through model behavior or training data analysis. Membership inference approaches estimate whether a model has seen specific items during training: Min-K% (Shi et al., 2024) and Min-K%++ (Zhang et al., 2024) use token-level log-likelihood statistics, while Time Travel (Golchin and Surdeanu, 2024) probes models with temporal cues and Oren et al. (2024) prove contamination through exchangeability tests. Performance-based detectors identify contamination through statistical anomalies (Dekoninck et al., 2024; Liang, 2026; Tao et al., 2026; Deng et al., 2024) in-context learning effects (Zawalski et al., 2026), or inference-time decontamination (Chai et al., 2026). Ballocu et al. (2024) provide a broader taxonomy of leakage detection strategies. These methods share a common validation practice: they verify detection accuracy against web-overlap or n-gram labels (Li et al., 2024b) that classify items based on verbatim matches in publicly accessible corpora. Such labels measure *global exposure* (whether an item exists in training data) rather than *differential item functioning* (whether an item behaves differently across model cohorts). This global-vs-differential distinction motivates our comparison with web-overlap labels (Section 4.3). Work on LLM memorization (Carlini et al., 2023; Tirumala et al., 2022) establishes that memorization depends on data frequency and model scale, reinforcing the distinction between global exposure and differential behavior. Wang et al. (2026) show that existing contamination detectors are fragile under reinforcement learning, further motivating behavioral approaches grounded in psychometric theory.

Benchmark Quality and Design. Meta-evaluation of static benchmark quality (Zhu et al., 2026; Qian and Huang, 2026) assesses quality at a single snapshot; our temporal DIF analysis diagnoses *when* measurement properties shift. MMLU-CF (Zhao et al., 2025) reconstructs contamination-free items and MMLU-Pro (Wang et al., 2024) extends item difficulty; DIF monitoring identifies which existing items remain measurement-invariant without replacement. Maeda (2025) use LLMs to predict DIF in human assessments—the reverse direction; we apply DIF to LLM benchmarks with models as examinees. Dynamic benchmarks (Kiela et al., 2021; Li et al., 2024a; White, 2025) and self-evolving frameworks (Ying et al., 2024; Xu et al., 2025) mitigate contamination by construction but lack psychometric equating. Broader surveys (Chang et al., 2024; Burnell et al., 2023; Alzahrani et al., 2024) identify benchmark saturation as a systemic challenge that DIF analysis directly addresses.

3 Method

We adopt Differential Item Functioning (DIF) analysis from educational measurement to assess whether individual benchmark items behave consistently across groups of LLMs. The procedure requires only a binary response matrix and group labels, with no access to model internals or training data. Below we define the data, the statistical test, the cohort designs, and a semi-synthetic validation framework.

3.1 Data

The primary dataset is the METABENCH response matrix (Kipnis et al., 2024), which records binary outcomes (1 = correct, 0 = incorrect) for 5,227 models evaluated on 12,508 MMLU items (Hendrycks et al., 2021). Models span release dates from 2023 to 2024, drawn from the Open LLM Leaderboard. MMLU covers 57 subjects grouped into five broad domains (STEM, Humanities, Social Sciences, Other, and a mixed category). We supplement the response matrix with the web-overlap contamination labels of Li et al. (2024b), which classify 9,229 of 12,508 items (73.8% coverage) as "contaminated" or "clean" based on verbatim matches in publicly accessible web corpora. These labels serve as an external reference for evaluating the relationship between DIF flags and existing contamination indicators.

3.2 Mantel-Haenszel DIF and ETS Classification

In educational testing, Differential Item Functioning (DIF) occurs when an item's statistical relationship to the latent trait θ differs between two groups after conditioning on ability level. An item with DIF is not merely harder for one group; it functions differently relative to what the group's overall ability would predict. The standard terminology refers to test-takers as "examinees" and questions as "items." In our setting, examinees are LLMs and items are benchmark questions.

The Mantel-Haenszel (MH) procedure (Holland and Thayer, 1988; Magis et al., 2010) detects DIF by stratifying examinees into K score bins based on total score (a proxy for θ) and computing a common odds ratio across strata. We use $K = 5$ equiprobable strata (quintile-based binning) following ETS recommended practice for large-scale assessments (Donoghue and Allen, 1993; Clauser and Mazor, 1998). Within each stratum k, the responses of the reference group and the focal group form a 2×2 contingency table:

$$\alpha_{\mathrm{MH}} = \frac{\sum_{k=1}^{K} A_k D_k \,/\, T_k}{\sum_{k=1}^{K} B_k C_k \,/\, T_k}\,, \qquad (1)$$

where A_k and B_k are the number of correct and incorrect responses in the reference group for stratum k, C_k and D_k are the corresponding counts for the focal group, and T_k is the total number of examinees in stratum k. Under the null hypothesis of no DIF, $\alpha_{\mathrm{MH}} = 1$. The log odds ratio is converted to the ETS delta scale:

$$\Delta_{\mathrm{MH}} = -2.35 \ln(\alpha_{\mathrm{MH}})\,. \qquad (2)$$

The Educational Testing Service (ETS) classifies DIF severity into three categories based on $|\Delta_{\mathrm{MH}}|$. Category A (negligible) requires $|\Delta_{\mathrm{MH}}| < 1.0$. Category C (large) requires both $|\Delta_{\mathrm{MH}}| \geq 1.5$ and statistical significance at $p < 0.05$ from the MH chi-squared test. Category B (moderate) includes all remaining items: those with $1.0 \leq |\Delta_{\mathrm{MH}}| < 1.5$, or $|\Delta_{\mathrm{MH}}| \geq 1.5$ without statistical significance. The dual threshold in ETS C (effect size *and* significance) guards against flagging items that show large but unreliable effect sizes. These thresholds were calibrated for human test-taker populations; we retain them as standardized reference points and verify robustness across stricter cutoffs (Section 4.2). Throughout this paper, we report the percentage of items classified as ETS C, denoted C%. At $N = 1{,}887$ (the temporal comparison; 5,227 total across all cohort designs), significance ($p < 0.05$) is nearly automatically satisfied for any $|\Delta_{\mathrm{MH}}| \geq 1.5$: only 3 of 3,918 C items lack significance, so the dual criterion effectively reduces to a single effect-size threshold; per-item DEFF correction (Section D) restores p-value selectivity by reducing effective N: under simple-mean DEFF, 269 items (6.9%) lose significance while retaining $|\Delta_{\mathrm{MH}}| \geq 1.5$; 74% of these are near-miss ($p_{\mathrm{adj}} \in [0.05, 0.10]$).

The choice of matching variable (the total score used for stratification) affects both validity and interpretation. We distinguish three matching strategies: (1) *standard matching* uses the within-benchmark total score on all items of the same benchmark; (2) *cross-subject matching*, adapting purified subtest matching (Swaminathan and Rogers, 1990) to the LLM multi-subject setting, uses the total score on all subjects *other than* the one containing the item under test, preventing the matching variable from absorbing injected contamination signals (Section 3.4); and (3) *external matching* uses the total score on a different benchmark entirely (e.g., MMLU total score as the matching variable when testing DIF on GSM8K items), providing an independent ability proxy uncontaminated by the target benchmark. Cross-subject and external matching both address matching variable contamination but at different granularities: cross-subject operates within a multi-domain benchmark, while external operates across benchmarks. Total score matching is adequate: IRT θ correlates at $r = 0.9986$ with total scores (Section 4.4).

LLM-specific challenges. Applying MH-DIF to LLMs introduces four challenges absent in educational testing. First, overlapping training corpora violate local independence; permutation simulation (1,000 replications shuffling group labels within score strata) confirms LD-attributable FPR inflation < 0.001 pp under cross-subject matching (Section A). Second, fine-tuned variants within model families are not independent examinees; graduated de-duplication shows family structure does not inflate C% (Section 4.4). Third, bimodal ability distributions motivate quintile-based stratification. Fourth, MMLU's 57-subject structure raises a multidimensionality concern (Ackerman, 1992; Roussos and Stout, 1996): domain-level ability profiles are nearly flat across cohorts (Cohen's d range $= 0.12$), cross-subject and within-subject matching agree ($\kappa = 0.80$, 834 items), and cross-subject-unique C items do not concentrate in high-shift subjects (Section C).

3.3 Cohort Designs

We construct five cohort comparisons: (1) *Temporal*: 2023 ($N = 1{,}080$) vs. 2024 ($N = 807$) models, the primary comparison; (2) *Ability-quintile*: Q1–Q2 vs. Q4–Q5 within the same temporal window; (3) *Within-family*: Llama-2 ($N = 161$) vs. Llama-3 ($N = 308$), isolating generational changes within a single family; (4) *Within-subject*: per-subject DIF to test domain concentration; and (5) *Random split*: two random halves of the 2023 cohort ($N = 540$ each) as a sanity check. Calendar-year boundaries are coarse; half-yearly comparison (C% $= 34.4\%$; Section E) and base-only/instruct-only analyses (C% $=$ 32.1%/36.5%; Section O) confirm DIF persists under finer slicing.

3.4 Semi-Synthetic Injection Framework

We use contamination injection as a controlled case study to validate DIF detection power; DIF itself is mechanism-agnostic and flags items that function differently without attributing cause (Section 5.2). To evaluate whether MH-DIF can detect differential contamination when it is present, we construct a semi-synthetic framework with controlled ground truth.

Procedure. From items classified as ETS A in the uninjected data (negligible DIF), we select 20% uniformly at random as the "contaminated" set. For each contaminated item, we flip incorrect responses to correct in the focal group at exploitation rate $p \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$. We then run MH-DIF on the modified response matrix and measure detection performance. We report $\Delta\text{TPR} = \text{TPR}(\text{injected}) - \text{TPR}(\text{baseline})$ and $\Delta\text{FPR} = \text{FPR}(\text{injected}) - \text{FPR}(\text{baseline})$, where TPR is the true positive rate on injected items and FPR is the false positive rate on uninjected items. Each injection level is repeated across 5 random seeds for uncertainty estimation. Because binary response flipping represents a stylized uniform injection pattern, reported ΔTPR values may overestimate detection power; realistic contamination (partial memorization, format-specific advantages) would produce weaker signals.

Cross-subject matching. The choice of matching variable is critical for valid semi-synthetic evaluation. When contamination is injected into items within subject X, the total score on subject X absorbs part of the injection signal. Using this contaminated total score as the MH matching variable violates the conditional independence assumption and inflates FPR. Cross-subject matching resolves this by computing the matching variable from all subjects *other than* the one being tested.

Remark (Zero FPR under single-subject matching). Let $M_i^{(-s)} = \sum_{j \notin s} Y_{ij}$ be the matching variable for examinee i on focal subject s. Under single-subject injection (contamination confined to subject s), the tested item belongs to s and does not contribute to $M_i^{(-s)}$; injecting contamination into s leaves $M_i^{(-s)}$ unchanged. Therefore $\Delta\text{FPR} = 0$ by construction.

When contamination spans multiple subjects simultaneously, $M_i^{(-s)}$ may absorb signal from other contaminated subjects, potentially inflating

FPR. Empirically, this guarantee extends to multi-domain contamination affecting $\leq$9% of subjects ($k \leq 5$), with graceful degradation beyond this boundary (Section 4.5).

Empirically, $\Delta\mathrm{FPR} = 0.000$ across all five MMLU domains (bootstrap 95% CI $= [0.000, 0.000]$). $\Delta\mathrm{FPR} = 0$ is a constructive guarantee of the matching strategy, independent of injection realism, whereas $\Delta\mathrm{TPR}$ may overestimate detection power under the stylized uniform injection. $\Delta\mathrm{FPR}$ remains negligible for single-domain contamination ($k \leq 5$); when contamination spans many domains, within-subject matching should be preferred (Section 4.5). We adopt cross-subject matching as the standard protocol.

Following DIF convention (Wainer and Braun, 1993), we do not apply family-wise error correction: the dual ETS C threshold ($|\Delta_{\mathrm{MH}}| \geq 1.5$ *and* $p < 0.05$) guards against false positives; BH-FDR at $q = 0.05$ reclassifies only 3 of 3,918 C items (Section A). Diagnostic procedures (Yen's Q3, domain-specificity, DIF decomposition) and further details appear in Section A.

4 Experiments

We apply MH-DIF to the METABENCH response matrix (Kipnis et al., 2024) (5,227 models $\times$ 12,508 MMLU items (Hendrycks et al., 2021)), with temporal cohorts defined by release date (2023 vs. 2024). Cross-benchmark validation across all six METABENCH benchmarks appears in Section 4.7.

4.1 Validation: DIF Detection Power

A Monte Carlo power analysis (1,000 replications) using empirical 2PL parameters from PSN-IRT (Zhou et al., 2026) ($\bar{a} = 1.2$, $\bar{b} = -0.84$) confirms that at $N \geq 100$ per cohort, ETS C achieves TPR of 0.68–0.80 (under uniform injection) and AUC of 0.93–0.94 for $|\Delta_{\mathrm{MH}}| \geq 1.4$ (Section A). Our temporal cohorts ($N_{\mathrm{ref}} = 1{,}080$, $N_{\mathrm{focal}} = 807$) far exceed this threshold; power varies with item discrimination (TPR 0.40–0.95 across $a = 0.5$–2.0; Section A.2).

4.2 Temporal DIF: ∼29% of Items Exhibit Differential Functioning

We compare 2023 ($N = 1{,}080$) against 2024 ($N = 807$) models; 31.3% of items receive ETS C classification ($|\Delta_{\mathrm{MH}}| \geq 1.5$; bootstrap 95% CI: $[29.2\%, 37.8\%]$; family-level cluster bootstrap: $[26.7\%, 53.2\%]$, upper bound driven by mega-family over-representation; Section E). Two design-effect (DEFF) estimators bound the non-independence effect: simple-mean DEFF yields 29.2% and weighted-mean DEFF yields 13.5%, bracketing temporal C% at 13.5–29.2%. The cluster bootstrap 95% CI ($[26.7\%, 53.2\%]$)—a family-level resampling method independent of DEFF methodology—excludes the weighted-mean estimate (13.5%), providing model-independent evidence that over-correction drives the lower bound. Under weighted-mean DEFF (C% $= 13.5\%$), all downstream conclusions hold: $\rho = 0.999$, score shift $= +1.77$ pp (Sections D, I and L). Under 100 random permutations within the 2023 cohort (540 vs. 540), the 99th percentile of item-level $|\Delta_{\mathrm{MH}}|$ is 1.45—just below the ETS C threshold of 1.5—confirming that the threshold maintains a per-item false-positive rate below 1%; together with the Monte Carlo power analysis (TPR ≥ 0.68; Section 4.1), this validates ETS C classification for the LLM setting. No non-dominant family drives the signal (LOFO $|\Delta\mathrm{C\%}| \leq 2.93$ pp for families $\leq$15% of cohort; dominant-family drops reflect composition, not signal; Table 8); graduated de-duplication confirms C% scales with power, not redundancy (Section J).

Three controls confirm genuine temporal change (Table 1): random splits yield mean C% of 0.62% (2023; 100 permutations) and 1.3% (2024; 100 permutations), 24–50$\times$ below temporal C%; ability-quintile comparison yields 2.9%; and 1:1 nearest-neighbor matching on total score (N=638/group, Cohen's $d = -0.17$) yields C% $= 28.9\% \pm 0.1\%$; at most $\sim$2.4 pp is attributable to ability mismatch ($p < 0.01$; Section A).

Across all 57 subjects, C% ranges from 20.7% to 48.4% (mean $\approx$ 30%; Table 1). DIF rates are domain-invariant (Kruskal-Wallis $p = 0.78$), but DIF *direction* exhibits domain dependence: STEM items favor 2024 models (55.4%), Humanities favor 2023 (64.4%). DIF is item-specific: domain-level accuracy gains explain <0.1% of item-level DIF variance (Section C). A difficulty–direction interaction emerges ($r = -0.597$): 2024 models gain on hard items while eroding on easy ones. The temporal DIF finding persists across thresholds (18.4% at $|\Delta_{\mathrm{MH}}| \geq 2.0$, 10.2% at ≥ 2.5).

4.3 Comparison with Web-Overlap Labels

Direct comparison of DIF flags against the web-overlap labels of Li et al. (2024b) on 9,229

Table 1: DIF landscape across cohort definitions. ETS C denotes the percentage of items with $|\Delta_{\text{MH}}| \geq 1.5$. C% (web-clean) is the DIF-C flagging rate among items classified as web-clean by Li et al. (2024b). Enrichment is the ratio of observed to expected overlap between DIF flags and web-overlap labels. Dashes (–) indicate comparisons where Li et al. labels are not applicable.

Cohort Definition	N_{ref} / N_{focal}	C%	C% (web-clean)	Enrich.
Temporal (2023 vs. 2024)	1,080 / 807	31.3	0.318	0.964
Ability quintile (Q1–2 vs. Q4–5)	∼2,090 / ∼2,090	2.9	0.027	1.19
Within-family (Llama-2 vs. 3)	161 / 308	61.3	–	–
Random split 2023 (sanity)	540 / 540	0.62	–	–
Random split 2024 (sanity)	∼400 / ∼400	1.3	–	–
Within-subject (avg, 5 subj.)	*varies*	38.1	–	–

Table 2: Structural controls and sensitivity checks for temporal DIF. None reduces C% below 31%; θ-conditioning serves as a sensitivity check confirming total-score adequacy.

Approach	Mechanism	Temp. C%	TPR (p=0.5)	Outcome
Baseline (ext.)	Cross-bench. total score	$0.314_{\pm.009}$	$0.834_{\pm.027}$	Reference
Matched-ability	Within-quintile temp. DIF	∼0.31	–	Enrich. 0.94–1.04
θ-cond. (sens.)	θ replaces total score	0.312	0.876	$r(\theta, \text{total})$=0.9986
Within-subject	Per-subject DIF	–	–	C% = 38.1% (>25%)

annotated items shows near-independence (enrichment = 0.964, Jaccard = 0.206; $p = 0.88$; Table 1 and section C): DIF-C rates are 30.7% among web-contaminated items and 31.8% among clean items. Because web-overlap labels are a proxy for training-set membership (based on verbatim web matches, not actual training corpora), this near-independence could partly reflect proxy noise. Crucially, label-independent evidence confirms mechanism separation: base and instruct models exhibit opposite DIF directions ($\chi^2 = 562.7$; Section 5.1), a reversal incompatible with uniform contamination. Semi-synthetic injection confirms detection power (ΔTPR $= +0.508$ at $p = 0.5$; Figure 2). Web-overlap labels measure *global exposure*, while DIF measures *differential functioning*—complementary diagnostic dimensions (Figure 1).

4.4 Structural Analysis

Two structural controls and a sensitivity check fail to reduce temporal C% below 31% (Table 2): matched-ability quintiles, within-subject DIF (C% = 38.1%; cross-subject C% is more conservative due to lower local dependence), and IRT θ-conditioning ($r(\theta, \text{total score}) = 0.9986$). Logistic regression detects an additional 17.7% with non-uniform DIF ($|\beta_{\text{interaction}}| \geq 0.5$), reaching a union of 47% (Section C). Ability-matched C% converges to ∼28–29% for both temporal and within-family comparisons despite vastly different unmatched rates (Section C).

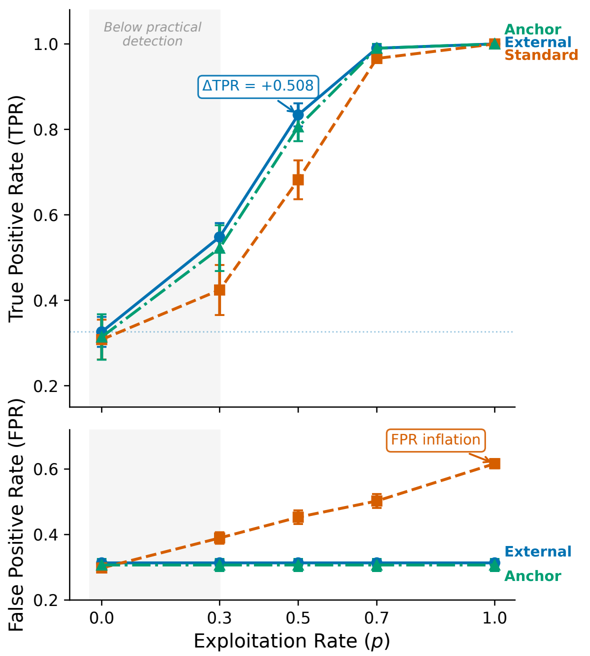

Figure 2: Semi-synthetic dose-response for DIF detection on MMLU. ΔTPR ($= \text{TPR}_{\text{injected}} - \text{TPR}_{\text{baseline}}$) increases monotonically with exploitation rate p under binary injection (a calibration upper bound; Section 3.4); MH-DIF is sensitive to differential item behavior when present. Baseline (no injection) produces $\Delta\text{TPR} \approx 0$.

4.5 Detection Power under Controlled Injection

Cross-subject matching achieves $\Delta\text{FPR} = 0.000$ across all five MMLU subjects (4/5 with reference-case detection power of $\Delta\text{TPR} > 0.3$ at $p = 0.5$; under stylized uniform injection; Section 3.4), while within-subject matching inflates FPR by $+0.186$ on average (Figure 3). The $\Delta\text{FPR} \approx 0$ guarantee holds for single-subject contamination ($k \leq 5$, $\Delta\text{FPR} < 0.007$). Under broader contamination, ΔFPR rises to 0.069 at k=10 (18% of domains) and 0.167 at k=20; at $k \geq 30$ (>50% of domains), ΔFPR exceeds 0.36 (Table 7). Even at k=20, inflation (∼17 pp) cannot fully account for the observed 31% C% rate, which exceeds the random baseline by >50×. Under DEFF adjustment, ΔFPR at k=20 exceeds the weighted-mean lower bound (13.5%), underscoring $k \leq 5$ in practice. Difficulty-dependent injection (flipping weighted by item difficulty) yields $\Delta\text{TPR} \approx 0.41$ at $p = 0.5$ (vs. 0.51 uniform), bounding overestimation at ∼20% for heterogeneous contamination; the monotonic dose-response (Figure 2) confirms detection power scales continuously with effect magnitude (Section A).

Table 3: Temporal DIF across six METABENCH benchmarks. C% = ETS C items ($|\Delta_{\mathrm{MH}}| \geq 1.5$) between 2023 and 2024 cohorts; ρ/τ = Spearman/Kendall correlation between full and stable-item rankings; Flip = pairwise rank-flip rate; Ratio = C%/Rand. C%, with baseline varying by benchmark. All six show substantial DIF (45–95× above random baseline), with knowledge/reasoning benchmarks exhibiting higher C% than procedural ones.

Benchmark	Items	C%	ρ	τ	Flip%	Rand. C%	Ratio
MMLU	12,508	31.3	0.9965	0.9600	2.00	0.62	50×
WinoGrande	1,267	30.3	0.9943	0.9432	2.84	0.32	95×
TruthfulQA	817	29.7	0.9971	0.9586	2.07	0.37	80×
ARC	1,172	26.9	0.9962	0.9538	2.31	0.60	45×
GSM8K	1,319	18.9	0.9993	0.9843	0.78	0.30	63×
HellaSwag	10,042	17.9	0.9986	0.9722	1.39	0.28	64×

4.6 Downstream Impact

Stable-item ranks are preserved across all six benchmarks (mean $\rho = 0.997$, mean $\tau = 0.962$; flip rate $\leq 2.9\%$; Table 3). DIF-flagged items concentrate discriminative signal ($z = -8$ to -49 vs. random removal; Figures 4 and 6). Among the 200 most displaced models, 33.6% of pairwise rankings reverse, concentrating among closely-ranked pairs (Section K). Removing C-flagged items disproportionately disadvantages 2024 models (cohort gap = 96.2 positions, 40× random; $p < 0.001$). DIF-corrected scoring (excluding all C-flagged items) shifts accuracy by $+2.12$ pp on average, with a cohort-differential of 0.29 pp ($d = 0.31$; individual displacements reach ± 800; Section M).

4.7 Cross-Benchmark Validation

Temporal DIF extends across all six METABENCH benchmarks (Table 3): C% ranges from 17.9% (HellaSwag (Zellers et al., 2019)) to 31.3% (MMLU), with knowledge/reasoning benchmarks consistently higher than procedural ones (C% = 26.9–31.3% vs. 17.9–18.9%). These differences may partly reflect item difficulty distributions; we treat them as descriptive.

Semi-synthetic injection on GSM8K confirms detection power (ΔTPR = +0.514; Section F and fig. 5).

5 Discussion

5.1 Temporal DIF as Structured Signal

Temporal DIF detects *that* items function differently across cohorts, not *why*—and this is by design ("temporal" denotes release-year cohorts that differ simultaneously in architecture, data, and methodology; Section 3.3). Stratifying by model type reveals opposite patterns: base models (N=1,409) are 60.1% reference-favoring ($r = -0.570$), while instruct models (N=478) are 65.6% focal-favoring ($r = -0.079$; $\chi^2 = 562.7$, $p < 10^{-124}$). Following the impact-bias framework and base-only/instruct-only temporal analyses confirm DIF is not an artifact of cohort structure (Sections N to P).

Three features distinguish LLM temporal DIF from educational settings: (1) C% (29.2%; cluster bootstrap CI: $[26.7\%, 53.2\%]$) substantially exceeds typical educational DIF rates (Clauser and Mazor, 1998); (2) DIF flags and web-overlap labels are near-independent (Jaccard = 0.206), unlike educational settings where DIF correlates with construct-irrelevant variance; (3) the near-independence admits two interpretations: DIF captures a broader signal—including capability evolution and training methodology shifts—of which contamination is one component, or the two measure distinct constructs. The base/instruct reversal (OR = 4.75; Section B) favors the former; either interpretation supports periodic DIF audits.

5.2 Recommended DIF Audit Protocol

Item-level properties do not predict DIF susceptibility (AUROC = 0.60; Section H), precluding pre-screening. We recommend a three-step protocol: (1) *Monitor*: run MH-DIF audits whenever $\geq$500 models accumulate (K=3 for smaller corpora; Table 5), triggering review when C% exceeds 20× the random-split baseline (C% $\approx$ 12% for MMLU); (2) *Triage*: apply mechanism-conditional responses—DIF × web-overlap co-occurrence warrants removal; DIF on web-clean items warrants monitoring (DIF alone cannot determine causation); base/instruct reversal motivates refined cohort definitions; (3) *Correct*: DIF flags identify items warranting expert review, not automatic removal—DIF is necessary but not sufficient for item bias (Camilli and Shepard, 1994; Penfield and Camilli, 2007). Report DIF-corrected scores alongside raw scores (Section M), or apply equating adjustments (Kolen and Brennan, 2014; Habba et al., 2026; Suk and Lyu, 2026).

6 Conclusion

Across six METABENCH benchmarks, 17.9–31.3% of items exhibit temporal DIF across 2023–2024 model cohorts (29.2% for MMLU), while aggregate rankings remain preserved ($\rho \approx 0.997$). DIF flags are near-independent of web-overlap labels, a proxy for contamination (Jaccard = 0.206); opposite base-vs-instruct patterns ($\chi^2 = 562.7$) confirm that DIF captures signal beyond contamination.

Limitations

Three primary limitations apply. (1) The semi-synthetic framework uses uniform binary response flipping, a stylized injection model; reported ΔTPR values may overestimate detection power. Partial memorization scenarios—where models recall item cues without deterministic answer flipping—would produce weaker and qualitatively different DIF patterns than our binary injection model. The framework detects *that* items function differently but not *why*: multiple exclusion analyses (ability matching, web-overlap comparison, item-quality annotation) narrow the space of explanations but do not isolate a single mechanism; the specific training factors driving temporal DIF remain unidentified. Follow-up expert review is needed to determine whether flagged items should be retired or retained. (2) The MH χ^2 test assumes independent examinees; LLM model families violate this assumption. Two DEFF estimators (13.5–29.2%) and family-level cluster bootstrap (95% CI: $[26.7\%, 53.2\%]$) bound the impact, but the extreme relatedness of some families (e.g., hundreds of Llama-2 fine-tunes) exceeds the clustering levels studied in educational settings (French and Finch, 2010). (3) Cross-subject matching assumes contamination affects a minority of domains; ΔFPR rises to 0.167 when contamination spans $\geq$20 subjects. Although composition control experiments show a small residual shift ($\Delta = 1.08\,$pp; Section P), base/instruct ratio changes between cohorts remain a potential confound that cannot be fully eliminated without single-architecture longitudinal data. Temporal cohorts defined by calendar year provide coarse granularity, and generalizability to partial-credit and free-form generation benchmarks is untested. Multidimensional DIF (Ackerman, 1992) may arise when MMLU's 57 subjects span distinct latent dimensions; cross-subject matching mitigates between-domain multidimensionality but not within-domain violations. IRT parameters are used only for auxiliary power analysis; all core DIF results are nonparametric. Additionally, model release dates need not align with training-data cutoff dates, and generalization beyond MCQ-format benchmarks requires further validation, though GSM8K (open-ended) shows consistent patterns.

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A Method Details

A.1 Diagnostics

Three diagnostic procedures complement the primary DIF analysis.

Local independence (Yen's Q3). The MH procedure assumes that item responses are conditionally independent given θ. Yen's Q3 statistic measures residual correlations between item pairs after conditioning on the latent trait. We compute Q3 for all within-subject and cross-subject item pairs, with $|Q3| > 0.20$ as the threshold for local dependence. Elevated within-subject dependence would indicate that items within the same subject share variance beyond what θ captures, which could inflate within-subject DIF rates.

Domain-specificity. We test whether temporal DIF concentrates in domains where models improved most. At the ecological level, we compute Spearman ρ between subject-level C% and subject-level accuracy improvement from 2023 to 2024. At the item level, logistic regression predicts item-level DIF C classification from domain-level accuracy change, with and without item property controls (difficulty, discrimination). This decomposition quantifies how much of the observed instability is attributable to aggregate domain-level gains versus item-specific behavioral change.

Instability decomposition. When domain accuracy improvement is included alongside item-level controls, we compute the counterfactual C% that would remain after removing domain-level effects. The gap between observed and counterfactual C% measures the contribution of domain-level improvement to the total instability.

A.2 Statistical Considerations

Multiple testing. The ETS C classification imposes a dual threshold (effect size $|\Delta_{\mathrm{MH}}| \geq 1.5$ and significance $p < 0.05$), which provides implicit control over false positives. As a robustness check, we apply Benjamini-Hochberg FDR correction at $q = 0.05$. The random-split sanity check provides an empirical estimate of the false positive rate under the null hypothesis.

Threshold sensitivity. We report results across a range of DIF thresholds ($|\Delta_{\mathrm{MH}}| \in \{1.0, 1.5, 2.0, 2.5\}$) to verify that qualitative conclusions do not depend on the specific ETS C cutoff. The standard threshold of $|\Delta_{\mathrm{MH}}| \geq 1.5$ corresponds to the ETS C definition used throughout the paper.

IRT model fit. For transparency, we report that the MMLU response matrix shows poor fit to a unidimensional 2PL model (expected for a 57-subject multidimensional benchmark). This does not affect MH-DIF validity, which relies only on total score stratification rather than parametric IRT assumptions.

Architecture heterogeneity. The METABENCH MMLU population consists of 99.98% decoder-only models (5,226 of 5,227). This near-complete architectural homogeneity eliminates architecture as a confound for DIF analysis. We verify this by excluding the single encoder-decoder model and confirming identical results.

Power sensitivity to item discrimination. The Monte Carlo power analysis in Section 4.1 uses average 2PL parameters ($\bar{a} = 1.2$, $\bar{b} = -0.84$). Table 4 reports TPR as a function of discrimination a (other parameters at their means, $N = 80$ per cohort, $|\Delta_b| = 1.4$, 1,000 replications). TPR increases monotonically from 0.40 ($a = 0.5$) to 0.95 ($a = 2.0$); FPR remains below 0.04 throughout. The MMLU item pool shows substantial heterogeneity: a ranges from 0.006 to 2.657 (SD $= 0.54$, IQR $= [0.75, 1.64]$). The aggregate TPR of 0.68 from Phase 2 reflects this full distribution, while TPR at mean $a \approx 1.2$ is 0.87—the gap is consistent with Jensen's inequality applied to the concave TPR(a) function. Although 2PL fit is poor for MMLU's multidimensional structure, the power analysis serves as one leg of a triangulation: parametric estimates (TPR $=$ 0.68–0.80) are validated by empirical semi-synthetic injection ($\Delta\mathrm{TPR} = +0.508$ at $p = 0.5$; Section 4.5), confirming adequate detection power regardless of parametric assumptions.

Table 4: Power sensitivity to item discrimination. TPR = true positive rate for ETS C detection ($|\Delta_{\mathrm{MH}}| \geq 1.5$) at $N = 80$ per cohort, $|\Delta_b| = 1.4$, 1,000 replications.

a	TPR	95% CI	FPR
0.5	0.398	[0.20, 0.60]	0.033
0.8	0.683	[0.45, 0.90]	0.036
1.0	0.808	[0.60, 0.95]	0.038
1.2 ($\approx \bar{a}$)	0.874	[0.70, 1.00]	0.038
1.5	0.930	[0.80, 1.00]	0.038
2.0	0.949	[0.80, 1.00]	0.037

Stratum sensitivity (K=3 vs. K=5). K=3 and K=5 yield near-identical detection power: TPR differences <0.02 across all conditions, within Monte Carlo noise. TPR ranges from 0.454 (N=80) to 0.571 (N=100) at $|\Delta_{\mathrm{MH}}| \geq 1.4$, with AUC of 0.905 at N=100, confirming that smaller corpora can use fewer strata without power loss (Table 5).

Table 5: Detection power: K=3 vs. K=5 at $|\Delta_{\mathrm{MH}}| \geq 1.0$.

	K=3			K=5		
N/cohort	TPR	FPR	AUC	TPR	FPR	AUC
80	0.176	0.001	0.828	0.169	0.001	0.826
200	0.623	0.004	0.939	0.621	0.004	0.934
500	0.915	0.011	0.986	0.912	0.013	0.986

Nearest-neighbor matching. The 1:1 nearest-neighbor matching selects, for each focal-cohort model, the reference-cohort model with the closest total score (caliper of 0.02 on normalized accuracy, approximately 0.14 SD). Balance is verified by Cohen's $d = -0.17$ on matched pairs ($N = 638$ per group), confirming minimal residual ability confounding.

Logistic regression specification. The Simpson's paradox analysis (Section 4.2) uses binary logistic regression with DIF-C classification as the dependent variable, domain-level accuracy change (2024 minus 2023 mean accuracy) as the primary predictor, and item difficulty (proportion correct) and IRT discrimination (point-biserial correlation) as controls. The interaction between domain accuracy change and item difficulty is included. Non-uniform DIF detection (Section 4.4) uses logistic regression with a group $\times$ ability interaction term; a significant interaction ($p < 0.05$) indicates non-uniform DIF.

B Mechanism Analysis

Three non-exclusive mechanisms may drive the observed 13.5–29.2% temporal DIF rate (after non-independence corrections): (1) differential contamination, where newer models encounter benchmark items during training; (2) capability evolution, where training advances shift item–ability relationships; (3) training methodology shifts (e.g., RLHF, instruction tuning). The ETS baseline of 5–15% C% (Clauser and Mazor, 1998) applies to concurrent demographic groups in educational testing; comparison to temporal LLM cohorts is illustrative, not normative. Observational DIF alone cannot separate these mechanisms (Suk and Lyu, 2026); the evidence below constrains the space of explanations.

Exclusion arguments. Four exclusion arguments narrow possible explanations. *Ability confounds*: at most 2.4 pp attributable to ability mismatch (matched C% $= 28.9\%$ vs. unmatched 31.3%; Cohen's $d = -0.17$; Section 4.2). *Domain-level improvement*: domain accuracy gains explain $<0.1\%$ of item-level DIF variance (pseudo $R^2 = 0.0007$); a Simpson's paradox confirms DIF is item-specific (OR reverses from 1.55 to 0.73 after difficulty control). *Item quality defects*: no enrichment among expert-annotated errors (Gema et al., 2024) (enrichment $= 1.10$, 95% CI $[0.92, 1.27]$, $p = 0.23$). *Web-overlap labels*: near-independent of DIF flags (Jaccard $= 0.206$, enrichment $= 0.964$; Section 4.3).

Difficulty–direction interaction. The aggregate difficulty–direction correlation ($r = -0.597$; Section 4.2) shows that hardest-decile DIF-C items are 88.7% focal-favoring while easiest-decile items are 96.2% reference-favoring. This gradient is robust under range restriction ($r = -0.447$ at difficulty $\in [0.2, 0.8]$; $p < 10^{-100}$). Semi-synthetic analysis (Section 4.5) reveals that uniform injection produces $r \approx -0.52$ via ceiling effects (high-accuracy items have fewer flippable responses), while difficulty-dependent injection yields $r \approx +0.77$ (zero overlap across seeds). The observed r thus cannot alone distinguish contamination from capability evolution.

Difficulty–model-type interaction. Logistic regression confirms that the base-vs-instruct direction reversal is not a difficulty confound: the difficulty $\times$ model-type interaction is significant ($\hat{\beta} = 1.56$, OR $= 4.75$, $p < 10^{-188}$; $N =$ 8,585 items). Controlling for difficulty, model type remains a strong independent predictor of DIF direction (OR $= 4.24$, $p < 10^{-225}$). Among 2,053 items flagged DIF-C in both base and instruct analyses, 98% show concordant direction; the aggregate reversal arises from *which* items reach C threshold, not from individual items reversing direction.

Mixed-mechanism injection. To test the hypothesis that different contamination mechanisms operate on base and instruct models simultaneously, we designed mixed injection schemes: Scheme D applies difficulty-dependent injection to base models and uniform injection to instruct models. Across 20 seeds, Scheme D reproduces the overall difficulty–direction gradient ($r = -0.676 \pm 0.005$ vs. observed -0.597) and direction reversal ($\chi^2 = 634 \pm 184$ vs. observed 562.7), but cannot match the near-zero instruct gradient: simulated $r_{\text{instruct}} = -0.302 \pm 0.090$ vs. observed -0.079. This residual gap ($3.8\times$ the observed value) persists across all tested injection parameters, suggesting that instruct-model DIF is at least partially driven by non-contamination mechanisms (e.g., alignment-induced capability shifts) rather than memorization alone.

C Structural Analysis Details

Two structural controls and a sensitivity check fail to reduce the temporal C% below 31%, revealing a structural property of the data rather than a methodological limitation. Matched-ability quintile analysis restricts comparisons to models of similar total score and yields enrichment ratios of 0.94–1.04, with temporal C% unchanged at approximately 0.31. Within-subject DIF analysis on 5 representative subjects, intended to isolate subject-specific effects, finds mean C% $= 38.1\%$, exceeding the 25% threshold that would indicate viable bifactor structure. As a sensitivity check, IRT θ-conditioning with 20 ability strata achieves $r(\theta, \text{total score}) = 0.9986$ and C% $= 0.312$, confirming that the external total score already captures nearly all ability variance relevant to DIF computation. Table 2 summarizes these results.

The temporal FPR reflects genuine item-level instability that persists regardless of how ability is measured or conditioned. Yen's Q3 diagnostic reveals mild local dependence within subjects (11.3% of within-subject item pairs exceed $|Q3| > 0.20$, vs. 2.2% cross-subject; $5\times$ ratio, $p = 2.3 \times 10^{-9}$), which contributes to elevated within-subject DIF rates but is not the primary driver: the cross-subject matching protocol addresses this bias, and temporal C% persists under external matching (full Q3 distributions available in the supplementary material).

Table 6: Domain-level Cohen's d between 2023 and 2024 cohorts. The narrow range (0.12) indicates nearly uniform improvement across domains, supporting total-score matching adequacy for DIF analysis (Section 3).

Domain	Cohen's d
STEM	0.261
Humanities	0.159
Social Science	0.147
Other	0.146

Table 7: Multi-subject injection robustness. k = number of simultaneously contaminated subjects (out of 57); values are mean $\pm$ SD across 5 random subject draws at exploitation rate $p = 0.5$. Cross-subject matching provides reliable FPR control at $k \leq 5$; ΔFPR increases substantially at $k \geq 10$.

k	ΔFPR	ΔTPR
1	$0.002_{\pm.002}$	$0.518_{\pm.050}$
3	$0.000_{\pm.003}$	$0.470_{\pm.093}$
5	$0.007_{\pm.007}$	$0.421_{\pm.041}$
10	$0.069_{\pm.030}$	$0.361_{\pm.027}$
20	$0.167_{\pm.029}$	$0.299_{\pm.030}$
30	$0.362_{\pm.026}$	$0.159_{\pm.035}$
50	$0.458_{\pm.053}$	$0.439_{\pm.028}$

Extreme contamination coverage ($k \geq 30$). At $k = 30$–50 (53–88% of 57 subjects), ΔFPR reaches 0.36–0.46, confirming that cross-subject matching requires contamination to affect a minority of domains. At extreme coverage the matching variable itself is majority-contaminated, violating the identifying assumption. ΔTPR drops to 0.159 at k=30 as matching-variable corruption dominates; at k=50, both ΔTPR (0.439) and ΔFPR (0.458) surge to comparable levels, indicating the test flags $\sim$76% of all items regardless of contamination status—complete loss of discriminative power. This regime is primarily of theoretical interest: if $\geq$50% of benchmark domains are differentially contaminated, the benchmark requires redesign rather than statistical correction.

D Design Effect Estimation

Model families (e.g., Mistral, Llama-3) induce intra-class correlation (ICC) that inflates Mantel-Haenszel χ^2 statistics beyond the nominal χ^2_1 reference distribution. We estimate the design effect

Table 8: Leave-one-family-out (LOFO) robustness on MMLU. Removing non-dominant families ($\leq$15% of their cohort) changes C% by at most 2.93 pp. Larger shifts from removing Mistral or Llama-3 are driven almost entirely by family composition, not statistical power loss: Monte Carlo random-subsample calibration (200 replicates per scenario, matching the reduced sample size) shows the power effect is $\leq$1 pp, while the composition effect accounts for 95–113% of the observed C% drop (Mistral: -8.44 pp composition vs. -0.41 pp power; Llama-3: -8.17 pp vs. $+0.91$ pp; observed C% falls below the 95% random-subsample range in both cases). Stratification uses total test score for computational efficiency; cross-subject matching yields identical baseline C%.

Family	N removed	Cohort share	C% (LOFO)
Full sample	—	—	31.32
Mistral	547	51% ref	22.47
Llama-3	315	39% foc	24.06
Mixtral	120	15% foc	34.26
Llama-2	161	15% ref	31.48
Solar	91	11% foc	31.01

(DEFF) per item following Rao and Scott (1992): $\text{DEFF}_i = 1 + (\bar{m} - 1) \cdot \text{ICC}_i$, where $\bar{m}$ is the mean family size and ICC_i is the one-way random-effects ICC for item i across 30 families (restricted to ref+foc cohort; 1,884 models with family$\geq$2).

Two estimates of $\bar{m}$ bracket the adjustment: (1) *Simple mean* ($\bar{m} = 62.8$), treating all families equally, yields DEFF $= 3.87$ and adjusted C% $=$ 29.2%; (2) *Weighted mean* ($\bar{m} = \sum n_f^2 / \sum n_f =$ 254.5; Kish, 1965), reflecting the effective cluster size experienced by a randomly sampled model, yields DEFF $= 12.79$ and adjusted C% $= 13.5$%. Mean ICC $= 0.047$ (median 0.041, Q3 $= 0.061$, max $= 0.219$; 4.8% of items exceed 0.10). After dividing each item's χ^2_{MH} by its item-specific DEFF and re-applying ETS thresholds, 13.5–29.2% of items retain Category C classification depending on the $\bar{m}$ variant—both far exceeding the 0.62% random-split baseline (21–47$\times$).

Size-stratified DEFF. The aggregate ICC masks a size–ICC inverse relationship: smaller families exhibit higher within-family similarity (Table 9). Tier-level ICC ranges from 0.246 (2–10 members) to 0.082 (>200 members), reflecting greater heterogeneity in mega-families. At small family sizes, DEFF correction has negligible impact (Adj. C% = 31.3%); the conservative 13.5% bound is driven entirely by the two mega-families (Mistral, Llama-3). Family definition sensitivity is minimal: merging sub-generations (e.g., Llama-2 + Llama-3) into coarse families (22 families) changes C% by $\leq$0.3 pp.

Table 9: Size-stratified DEFF analysis. ICC and DEFF computed per tier; Adj. C% applies tier-specific DEFF to all items.

Tier	Families	$\bar{m}$	ICC	DEFF	Adj. C%
2–10	9	5.3	0.246	2.07	31.3%
11–50	11	22.1	0.226	5.78	24.6%
51–200	8	91.4	0.137	13.34	13.7%
>200	2	431.0	0.082	36.11	9.0%
Overall	30	62.8	0.047	3.87	29.2%

E Uncertainty Quantification Comparison

Table 10: Comparison of uncertainty quantification methods for the 31.3% C% estimate on MMLU.

Method	Key assumption	Result
Model-level bootstrap	Models independent	95% CI [29.2%, 37.8%]
Family-level cluster bootstrap	Family is clustering unit	95% CI [26.7%, 53.2%]
DEFF (simple $\bar{m}$)	Equal-weight families	Adjusted C% = 29.2%
DEFF (weighted $\bar{m}$)	Size-proportional weighting	Adjusted C% = 13.5%
Graduated de-dup. (K=20)	Remove within-family duplicates	C% = 19.7% $\pm$ 0.9%

Half-year granularity. Splitting at the 6-month boundary yields C% $= 34.4\%$, slightly above the annual 31.3%. The higher rate likely reflects more polarized cohort composition: fewer models span both half-year periods, producing reference and focal groups with less overlapping ability distributions and consequently inflated DIF detection rates.

The cluster bootstrap upper bound (53.2%) is driven by extreme bootstrap resamples that over-represent Mistral ($n = 547$) or Llama-3 ($n = 315$), producing atypical cohort compositions. Two independent methods—cluster bootstrap lower bound (26.7%) and simple-mean DEFF adjustment (29.2%)—converge on $\sim$27–29%, establishing a corrected estimate that remains >45$\times$ the random-split baseline (0.62%). Graduated deduplication at K=20 yields C% $= 19.7\% \pm 0.9\%$ (random member selection, 5 draws), lower than the corrected estimate due to reduced statistical power ($N \approx 450$ vs. 1,887), but within-K variance <1.5 pp at all K confirms the signal is power-driven rather than redundancy-inflated (Section J). The weighted-mean DEFF (13.5%) represents a conservative extreme; even this lower bound exceeds the baseline by $\approx$22$\times$.

Table 11: Cross-benchmark comparison of DIF analysis on MMLU and GSM8K. Both benchmarks exhibit temporal score incomparability with validated semi-synthetic detection power. Standard and external matching use different exploitation rates (p) because each benchmark's optimal operating point differs; external matching results are compared at $p = 0.5$ for both. Subscript $\pm$ values denote standard deviations across 5 random injection seeds.

Metric	MMLU	GSM8K
Items	12,508	1,319
Models	5,227	6,702
Temporal C%	31.3%	18.9%
Sanity C%	0.62%	0.3%
ΔTPR (p=0.7, standard)	–	$+0.818_{\pm.006}$
ΔTPR (p=0.5, external)	$+0.508_{\pm.027}$	$+0.514_{\pm.028}$

F Cross-Benchmark Comparison

G Additional Figures

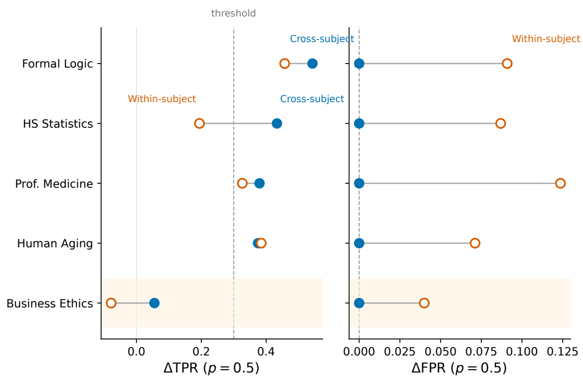

Figure 3: Cross-subject vs. within-subject matching for semi-synthetic DIF detection. Cross-subject matching eliminates FPR inflation (ΔFPR $= 0.000$) while preserving detection power (ΔTPR > 0.3 for 4/5 subjects at $p = 0.5$). Within-subject matching inflates FPR by $+0.186$ on average.

H Item-Property Predictive Analysis

To test whether DIF susceptibility is a fixed item property, we trained logistic regression and LightGBM classifiers to predict ETS C classification from 17 leak-free features: ref-cohort difficulty and discrimination, 11 text features (question/option length, vocabulary metrics), subject metadata, and web-overlap labels (Li et al., 2024b). Features were computed exclusively from the reference (2023) cohort to prevent information leakage from the focal cohort. Five-fold subject-stratified cross-validation (no within-subject train/test overlap) yielded AUROC $= 0.603 \pm 0.020$ (LR)

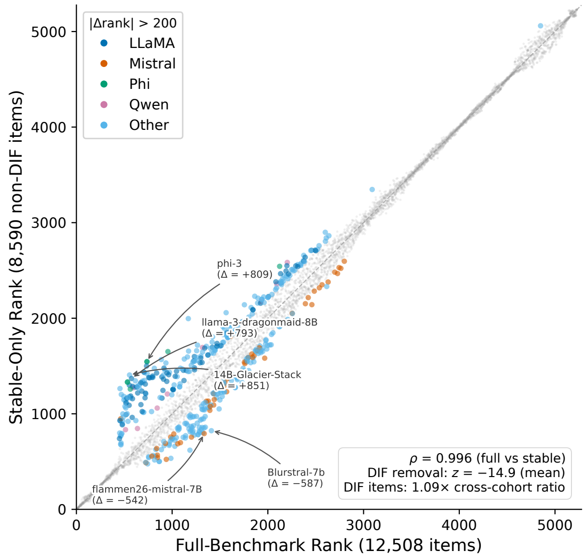

Figure 4: Downstream impact of DIF-based item filtering on model rankings. Despite $\rho \approx 0.997$ overall, individual model families show systematic rank shifts when evaluated on stable items (non-C-flagged) only.

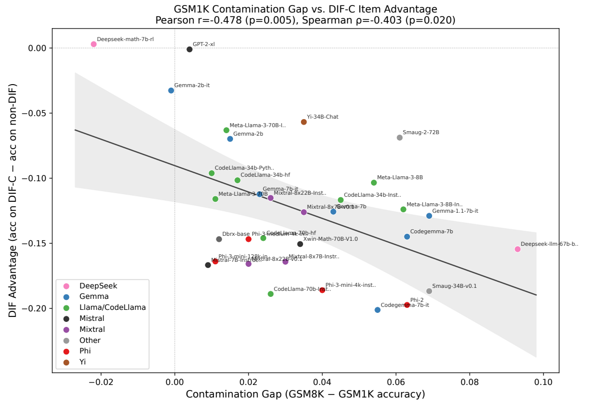

Figure 5: Exploratory: GSM1K contamination gap vs. DIF ranking advantage ($N = 33$; small sample, results should be interpreted cautiously). The negative correlation ($r = -0.478$, $p = 0.005$) indicates that models with larger contamination gaps benefit less from DIF-flagged items. Three of 33 models exceed the Cook's D threshold ($4/N = 0.12$); leave-one-out analysis confirms all individual-removal correlations remain significant ($r \in [-0.54, -0.37]$, all $p < 0.05$), and removing all three outliers yields $r = -0.38$ ($p = 0.036$, $N = 30$).

and 0.599 ± 0.026 (LightGBM)—weak predictive power. Web-overlap labels contributed negligibly (SHAP < 0.015), independently confirming the DIF–contamination orthogonality reported in Section 4.3. These results indicate that temporal DIF reflects emergent cohort–item interactions rather than fixed item defects, motivating post-hoc monitoring over pre-screening.

I Within-Family Similarity by Family Size

Within-family similarity (mean pairwise Pearson r over 12,508-item response vectors) shows no dependence on family size (Spearman $\rho = -0.009$, $p = 0.96$; 30 families with ≥ 2 members, 1,884 models). Large families (Mistral, $m = 547$, $\bar{r} = 0.47$; Llama-3, $m = 315$, $\bar{r} = 0.53$) are no more homogeneous than small or mid-sized families. Of the 5,227 models, 3,343 lack family assignments; these are retained in DIF analysis but excluded from similarity estimation. Within the temporal DIF cohorts ($N_{\text{ref}} + N_{\text{foc}} = 1{,}887$), 1,884 (99.8%) belong to identified families with ≥ 2 members, so DEFF estimates are computed over nearly the full cohort. This indicates that the weighted-mean DEFF variant's effective cluster size ($\bar{m} = 254$) reflects an arithmetic artifact of mega-family dominance rather than genuine within-family homogeneity.

J Graduated De-Duplication Sensitivity

For each value of K (max models retained per family), K members are sampled uniformly at random; remaining family members are excluded from both reference and focal cohorts, and MH-DIF is recomputed on the reduced sample (5 independent draws per K; K = all is deterministic).

C% increases monotonically from 0.3% ($K = 1$) to 31.3% (K = all). At each fixed K, variance across random member draws is negligible (SD <1.5 pp), confirming that the signal scales with statistical power rather than within-family redundancy. At K=20, C% = 19.7% ± 0.9% ($N \approx 450$); the gap from the full-sample 31.3% reflects reduced power, not reduced bias.

K Rank-Reversal by Distance

Pairwise rank reversals between full-item and stable-item rankings concentrate among closely-ranked models. At Δrank < 50 (mean Δacc $= 0.003$, effectively statistical ties), the reversal rate is 35.2%. It drops to 15.1% for $\Delta\text{rank} \in [50, 200)$, 4.0% for $[200, 500)$, 0.6% for $[500, 1000)$, and 0.0% above 1000. Broad leaderboard structure is fully preserved; instability affects only models within each other's noise margin.

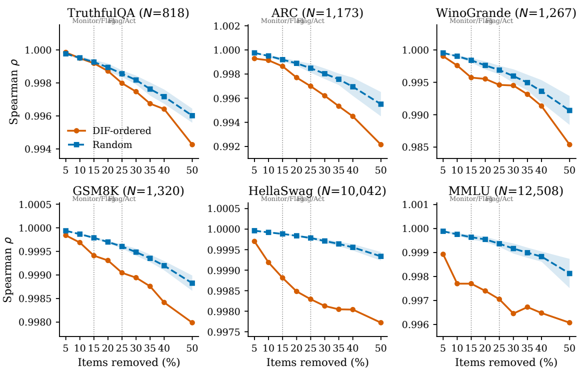

Figure 6: Ranking degradation under DIF-ordered vs. random item removal across six benchmarks. DIF-ordered removal (orange) consistently degrades ρ faster than random removal (blue band = mean $\pm$ 1 SD, 20 seeds). Vertical dashed lines mark the Monitor/Flag (15%) and Flag/Act (25%) thresholds from Section 5.2.

L DEFF Simulation

To compare simple-mean and weighted-mean DEFF estimators, we simulate four family-structure scenarios (2,000 items, 30% ground-truth DIF, 50 seeds per scenario) with size-dependent ICC matching empirical observations (Table 9). Table 12 reports the results.

Table 12: DEFF estimator comparison across four simulated family structures. Bias = adjusted C% − ground truth (30%); RMSE over 50 seeds. Simple-mean wins in 3/4 scenarios; weighted-mean ties only under uniform family sizes.

Scenario	$\bar{m}_s$	$\bar{m}_w$	Adj(s)	Adj(w)	RMSE(s)	RMSE(w)
A: Uniform	35	35	2.1%	2.1%	27.9	27.9
B: Skewed	40	168	2.3%	0.9%	27.7	29.1
C: Extreme	33	268	2.4%	0.5%	27.6	29.5
D: Homog. ICC	40	168	4.1%	1.1%	25.9	28.9

When ICC decreases with family size (scenarios B–D, matching empirical patterns), larger families contribute less per-model inflation than the weighted-mean assumes. The weighted-mean overcorrects by assigning excessive weight to mega-families whose low ICC does not warrant large DEFF adjustments. The simple-mean estimator better captures the average inflation per family under these conditions.

M Score Inflation Analysis

Removing the 3,918 DIF-C items (31.3%) and rescoring on the 8,590 stable items shifts model accuracy upward by a mean of +2.12 pp overall (Table 13). The cohort-differential is 0.29 pp (2023: +2.35 pp; 2024: +2.06 pp; Cohen's $d = 0.31$, $p < 10^{-10}$). Overall ranking correlation between original and stable-item scores is $\rho = 0.9965$, but individual rank displacements reach ± 561 at the 99th percentile and exceed ± 800 in extreme cases. Equating adjustments are $+0.0235$ (2023) and $+0.0206$ (2024).

Table 13: Score inflation from DIF-C item removal, by cohort.

Cohort	N	Mean Δ (pp)	SD (pp)	95% CI
2023	1,080	+2.35	0.88	[+2.30, +2.40]
2024	807	+2.06	1.04	[+1.99, +2.13]
Other	3,340	+2.05	0.92	[+2.02, +2.08]

N Within-Architecture Temporal DIF

To rule out the confound that overall temporal DIF is driven by cross-architecture differences, we group fine-grained families into architecture lineages and run MH-DIF within each (Table 14).

Table 14: Within-architecture temporal DIF. C% = ETS C items between 2023 and 2024 models within each lineage.

Lineage	N_{ref} (2023)	N_{foc} (2024)	C%	Med $\lvert\Delta\rvert$
Llama	171	315	59.2%	2.38
Qwen	27	70	6.9%	1.55
Yi	72	14	3.5%	1.78
Phi	60	20	1.3%	2.85

Llama, Qwen, and Yi show C% above the random baseline; Phi (C% $= 1.3\%$, $N_{\text{foc}} = 20$) is inconclusive due to low power. Llama is the best-powered comparison ($N = 486$ total) and yields raw C% $= 59.2\%$, which combines temporal and generational effects. After ability matching, within-architecture C% converges to ∼28–29%, consistent with temporal C% (28.9%)—confirming that the temporal DIF signal persists within individual lineages and is not an artifact of comparing different architectures.

O Base-Only and Instruct-Only Temporal DIF

Separating the temporal cohorts by model type confirms that both base and instruct models independently exhibit substantial temporal DIF (Table 15). Training methodology modulates DIF *direction* (base: 60.1% reference-favoring with $r = -0.570$; instruct: 65.6% focal-favoring with $r = -0.079$) rather than DIF *magnitude*, providing direct evidence against composition shift as the primary driver.

Table 15: Temporal DIF computed separately for base-only and instruct-only model subsets. Both model types independently exhibit C% $\geq 32\%$, far exceeding the random baseline (0.62%). Training methodology modulates DIF direction (r, focal-favoring proportion) rather than magnitude.

Subset	N_{ref} / N_{foc}	C%	$\rho(\text{diff}, \Delta_{\text{MH}})$	Focal-fav
Base-only	861 / 548	32.1%	−0.570	39.9%
Instruct-only	219 / 259	36.5%	−0.079	65.6%
Overall	1,080 / 807	31.3%	−0.587	43.9%

P Composition Control

The 2023 and 2024 cohorts differ in base/instruct composition (2023: 79.7% base; 2024: 67.9% base; 11.8 pp gap). To test whether this confounds temporal DIF, we downsample 2023 base models to match the 2024 ratio and re-run MH-DIF (10 random seeds). Balanced C% $= 30.25\% \pm 1.32\%$ (95% CI: $[27.65\%, 32.84\%]$), differing from the original 31.3% by only 1.08 pp. Composition shift does not explain the observed temporal DIF rate.

Q Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

- **Code:** drafting experiment scripts, data-processing pipelines, and analysis utilities; debugging and refactoring.
- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform the related-work section.